

MUNEEB UR REHMAN

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I am a **final-year Artificial Intelligence student** with hands-on experience building and deploying end-to-end AI applications across Computer Vision, Natural Language Processing, and Big Data Analytics. Skilled across the entire machine learning lifecycle, from engineering efficient data pipelines and fine-tuning PyTorch models to deploying containerized microservices with FastAPI and Docker. I combine these skills to develop practical AI solutions that translate research concepts into reliable, user-facing applications while focusing on efficient software design and performance optimization.

Projects

Zero-Shot Character Detection & Tracking | Metric Learning, Multi-Object Tracking, Inference Optimization

- Developed a zero-shot anime character detection and tracking pipeline using fine-tuned DEIMv2, LoRA-adapted DINOv3, and vector similarity search for character re-identification across images and videos without retraining.
- Optimized and deployed the pipeline with ONNX, achieving near real-time GPU inference by processing 1 second of video in approximately 2.5 seconds.

BU Chatbot | RAG, LLMs, FastAPI, Vector Databases, JWT Authentication, Cloud Deployment, Performance Optimization

- Developed a RAG-based AI chatbot for the Bahria University Student Rulebook using LangChain, MongoDB Atlas Vector Search, and Groq LLMs, converting static documents into an interactive assistant with conversational memory and secure authentication via Clerk.
- Containerized and deployed the backend on Azure with a Vercel-hosted frontend, reducing response latency from 55s to 5s.

NYC Fare Predictor | SQL, EDA, Big Data Analysis, Business Rule Engineering, Ensemble ML Models, Hugging Face Spaces

- Developed a hybrid NYC taxi fare prediction system using 40M+ trip records, combining five ML models for variable fare components with EDA-driven business rules for fixed charges.
- Achieved 8.5% average prediction error and deployed an optimized CPU inference pipeline on Hugging Face Spaces for fast real-time predictions.

Technical Skills

- AI/ML:** Deep Learning, Computer Vision, NLP, Transformers (LLMs/VLMs/ViTs), Transfer Learning, Retrieval-Augmented Generation (RAG), Parameter-Efficient Fine-Tuning (LoRA/QLoRA), Generative AI
- Data & Deployment:** Linux, WSL, Hugging Face, MongoDB Atlas, ONNX, Git/GitHub Actions, Docker, Azure, Vercel
- Data Tools:** Google AppSheet, Excel, SQL, Microsoft SQL Server, Power BI, Apache Parquet
- Programming Languages:** Python, C++, Java

Certifications

- Advanced Data Analytics - Google
- Model Context Protocol - Anthropic
- DevOps to MLOps Bootcamp - Packt
- Deep Learning with Keras and TensorFlow - IBM

Education

BS in Artificial Intelligence | Bahria University | Karachi Campus

- Currently in 7th semester with 3.3/4.0 CGPA
- Co-founded and led the ACCP Club Arts Team, managing a 20-member creative team responsible for event branding and artwork recognized by university directors and featured at official events.

Intermediate in Pre-Engineering | Bahria College Karsaz | Karachi Board

- Graduated with Grade A
- Served as Captain of the Table Tennis Team for the House in the Interhouse League, leading the team to a runner-up finish.